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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-18 (Makeup)

Electronics and Communication Engineering

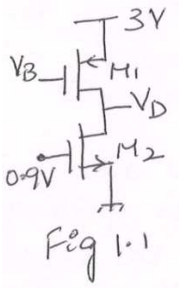
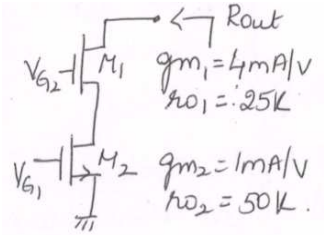
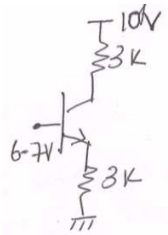
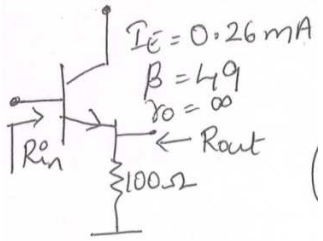
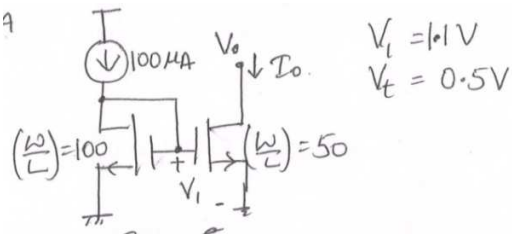
ANALOG MICROELECTRONIC CIRCUITS

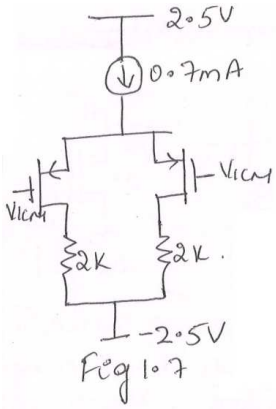
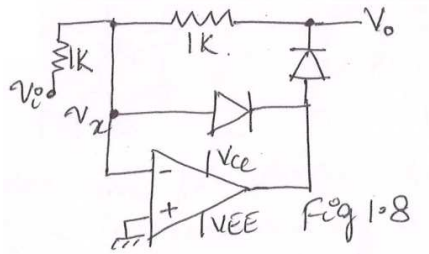
Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

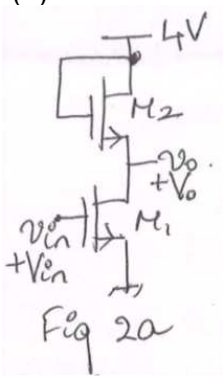
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

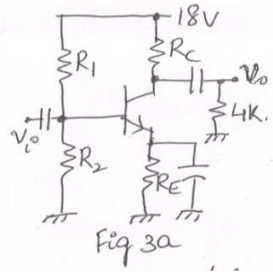
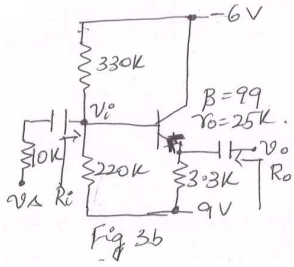
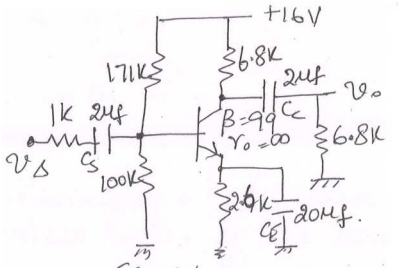
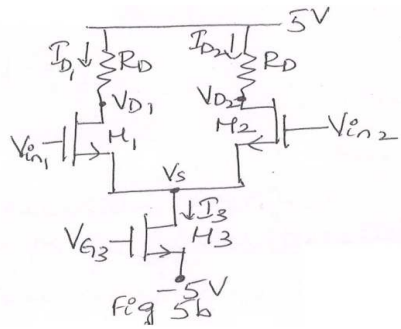
PART-A

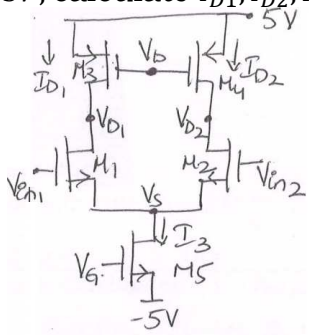
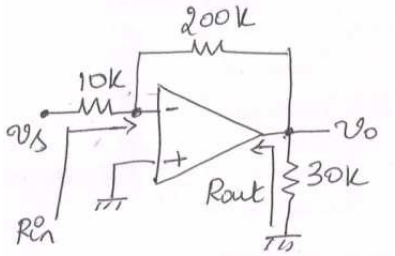
1	1.1	Given $K_n'\left(\frac{W}{L}\right)_n = 4K_p'\left(\frac{W}{L}\right)_p$, $V_{tn} = -V_{tp} = 0.5V$ and $\lambda = 0$; calculate the range of voltage that V_D can take, so that $M1$ and $M2$ as shown in Fig 1.1 are in saturation.  Fig 1.1  Fig 1.2  Fig 1.3	02
	1.2	The output impedance R_0 in the circuit of Fig 1.2 is approximately _____.	02
	1.3	In the circuit of Fig 1.3, assuming the transistor has a large β, V_{CE} = _____ and I_C = _____.	02
	1.4	The input impedance, R_{in} in the circuit of the Fig 1.4 = _____.  Fig 1.4  Fig 1.6	02
	1.5	The output impedance, R_{out} in the circuit of the Fig 1.4 = _____.	02
	1.6	In the current mirror circuit of Fig.1.6, I_o = _____ and $V_{0_{Min}}$ = _____.	02

1.7	In the MOS differential pair shown in Fig. 1.7, $V_{1cm(\min)} =$ _____ and $V_{1cm(max)} =$ _____.  $Kp'(\frac{W}{L}) = 3.5mA/V^2$ $V_t = -0.8V$ $V_{CS(min)} = 0.5V.$ Fig 1.7  Fig 1.8	02
1.8	For the circuit shown in Fig 1.8, assume the OP amp is ideal with $V_{CC} = -V_{EE} = 5V$. The diodes are ideal. a) Given $V_i = 1V; V_0 =$ _____, $V_X =$ _____. b) Given $V_i = -1V; V_0 =$ _____, $V_X =$ _____.	02
1.9	Current shunt feedback is provided to an amplifier to _____ its output impedance and _____ its input impedance.	02
1.10	An voltage amplifier needs $50mV$ input to give a certain output. When negative feedback is provided to this amplifier, it needs $2.5V$ to deliver the same output. If the closed loop gain of the amplifier is $38dB$, its open loop gain feedback factor are _____ and _____ respectively.	02

PART-B

2	a	In Fig 2a, the transistors have $\mu_n C_{ox} = 0.4\frac{mA}{V^2}, V_t = 0.5V$, $\lambda = \gamma = 0, \left(\frac{W}{L}\right)_1 = 10$ and $\left(\frac{W}{L}\right)_2 = 2.5$ i) If $V_{in} = 0.6V$, calculate the small signal voltage gain. ii) At what input voltage $M1$ delivers $0.1V$ into triode region (i,e $V_0 = V_{in} - V_t - 0.1$)? iii) Calculate the drain currents of $M1$ and $M2$ under the condition stated in (ii)  Fig 2a	04 04 02
	b	Derive an expression for small signal voltage gain and output impedance of a $NMOS$ source follower. Assume $\lambda \neq 0$.	06

3	a	Design the values R_C, R_E, R_2 and R_1 in the CE amplifier circuit of Fig 3a to obtain $V_{CE} = 6V, I_E = 1mA, S(I_{CO}) = 10$ and small signal voltage gain $\frac{V_0}{V_i} = -100$. The transistor has $\beta = 99$ and $r_o = \infty$.  Fig 3a  Fig 3b	08
	b	In Fig 3b, determine the operating point of Si PNP transistor, small signal voltage gain $\frac{V_0}{V_i}, \frac{V_0}{V_s}$, the input impedance R_i and the output impedance R_o. The transistor has $\beta = 99$ and $r_o = 25k\Omega$.	08
		OR	
4	a	Derive an expression for small signal voltage gain and output impedance of Darlington pair emitter follower.	08
	b	In the amplifier circuit of Fig.4b, determine the lower cut off frequencies F_{cs}, F_{Lc} and F_{CE} due to C_s, C_c and C_E respectively and the upper cut off frequencies, f_{hi} and f_{ho} due to the input and output capacities respectively. Assume $C_{be} = 50pf$ and $C_{be} = 5pf$ for the transistor.  Fig 4b.	08
5	a	Derive an expression for output impedance of a $NMOS\ WILSON$ current mirror.	04
	b	In Fig 5b, all transistors have $\mu_n C_{ox}\left(\frac{W}{L}\right)_{1,2} = 2mA/V^2 \mu_n C_{ox}\left(\frac{W}{L}\right)_3 = 4\frac{mA}{V^2}$, $V_t = 0.5V, V_{G3} = -3.5V, R_D = 3k\Omega$ and $\lambda = \gamma = 0$. Calculate $I_{D1}, I_{D2}, I_3, V_{D1}, V_{D2}$ and V_s for each of the following cases. i) $V_{in1} = V_{in2} = 0V$ ii) $V_{in1} = V_{in2} = 5V$ iii) $V_{in1} = 0V \quad V_{in2} = 0.1V$  Fig 5b	12

		OR	
6	a	Derive an expression for output impedance of a $BJT\ WILSON$ mirror.	04
	b	In Fig 6b, all transistors have $\mu_n C_{ox} = 0.4mA/V^2 \mu_p C_{ox} = 0.2\frac{mA}{V^2}$, $V_m = -V_{tp} = 0.5V$. Given $\left(\frac{W}{L}\right)_{1,2} = 10$ and $\left(\frac{W}{L}\right)_{3,4,5} = 20$, $\lambda = \gamma = 0$, $V_b = 3.5V$ and $V_G = -3.5V$. i) If $V_{in1} = V_{in2} = 0$, calculate I_{D1}, I_{D2}, I_3 and V_s. ii) If $V_{in1} = 0$ and $V_{in2} = 5V$, calculate $I_{D1}, I_{D2}, I_3, V_{D2}$ and V_s.  Fig 6b.	12
7	a	With the help of a neat circuit, explain the working of a full wave precision rectifier where none of the op-amps go into saturation.	08
	b	Design an instrumentation amplifier, using three ideal op-amps, to realize a differential gain, variable in the range 0.5 to 25, utilizing a $100k$ pot as a variable resistor.	08
8	a	The op amp in the circuit of Fig 8a has an input impedance of $50k\Omega$ and output impedance of $10\,k\Omega$ and an open loop gain of 10^4. Determine $\frac{V_o}{V_s}$, R_{in} and R_{out} using feedback theory.  Fig 8a.	10
	b	A class B complementary symmetry power output stage operates at $12V$ single power supply for the load of 4Ω and the input of $3sin\omega t$, determine the output power, input power, efficiency, dissipation in each transistor. Also calculate the power dissipation capability of each transistor.	06